

Ph.D. researcher in Computer Science specializing in applied machine learning for code understanding, automated refactoring, and empirical evaluation. Experienced with LLMs/transformers and NMT, large-scale experimental design (10M+ datapoints), and rigorous statistical analysis (Mann–Whitney U, Kendall’s  $\tau$ , mutual information, SHAP). Building production-style pipelines for LLM inference and multi-agent evaluation of refactorings; published benchmark and replication studies for program repair and text retrieval-based bug localization.

SKILLS

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|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ML/AI Techniques          | LLMs & Transformers (GPT, Qwen), Fine-tuning & Quantization, LLM Inference (vLLM), Statistical Modeling                           |
| ML/DL Frameworks          | PyTorch, Machine Learning (Scikit-learn), Deep Learning (PyTorch, TensorFlow, SciKeras), Hugging Face, RAPIDS cuML, NumPy, Pandas |
| Statistics & Data Science | Hypothesis Testing, Feature Engineering, Exploratory Data Analysis, A/B Testing                                                   |
| Software Engineering      | System Design, Performance Optimization, Code Refactoring, API Development                                                        |
| Programming Languages     | Python, Java, C/C++, SQL, Bash, LaTeX                                                                                             |
| Tools & Infrastructure    | Linux, Docker, Git, SLURM, Weights & Biases                                                                                       |

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

William & Mary

Sep 2022 - Present

Williamsburg, VA, USA

- Designing a multi-agent pipeline to evaluate LLM-generated code refactorings for semantic correctness and readability, integrating relative-comprehensibility models with differential testing and formal verification (in progress).
- Introduced a relative-comprehensibility (RC) measurement framework that improved prediction accuracy by 150% over a naive baseline and produced a 4.4× larger gain than state-of-the-art comprehensibility prediction models; evaluated across 10M+ datapoints using an 8-model ML pipeline (Random Forest, SVM, XGBoost, etc.) and two LLMs (GPT-5.4, Qwen-2.5) — a scale comparable to production recommendation/ranking system evaluation. Conducted a user study with developers (Qualtrics) to collect data and validate the usefulness of RC models (submitted to TOSEM).
- Designed and executed a controlled experiment (A/B Style 1,228 control and 935 treatment classifiers across 6 model families and multiple comprehensibility metrics) showing formal-verifier warning counts do not improve comprehensibility prediction; applied Mann–Whitney U (95% CI), Kendall’s  $\tau$ , mutual information, and SHAP analyses to validate results (EASE’26).
- Replicated RewardRepair, an NLP transformer-based model for program repair, across 960 generated patches for 96 UI-centric Android bugs (46 synthetic, 50 real); measured a 3.4% plausible patch rate and 0% correct patches for real bugs, demonstrating that repair techniques without UI-specific fine-tuning perform poorly on GUI-related issues (ISSTA’26).

Senior Software Engineer

Enactor Ltd (UK-based)

Feb 2020 - July 2022

Sri Lanka

- Improved platform maintainability by 20% across two large-scale production systems through systematic profiling, debugging, and refactoring, resolving key performance bottlenecks.
- Owned system design, development, deployment, testing, and performance tuning for React front-end and backend modules used by enterprise clients.
- Enhanced deployment environments as part of the Deployment & Scalability team, introducing Docker-based infrastructure improvements and automating environment provisioning with cloud infrastructure (AWS, Terraform, Ansible).
- Delivered ‘ShoeWall’, a customer-facing feature for Frasers Group; owned technical specification, solution design, code review, and automated test coverage using Selenium, TestNG, and JUnit.

- Collaborated in Agile environments to build JAX-RS REST APIs, database backend modules, and automated testing pipelines for Apache OODT 2.0; later served as an Apache OODT PMC Member & Committer (2021-2023).

EDUCATION

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|---------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>William &amp; Mary</b>                                                                   | Sep 2022 - Expected graduation May 2028 |
| <i>Ph.D. in Computer Science</i> (GPA: 4.00 / 4.00)   Advised by Prof. Oscar Chaparro       | Williamsburg, VA, USA                   |
| • <b>Achievements:</b> International Student Opportunity Scholarship (2024)                 |                                         |
| <b>University of Moratuwa</b>                                                               | Feb 2016 - Jan 2020                     |
| <i>B.S. in Computer Science &amp; Engineering</i> (GPA: 3.53 / 4.00)                        | Sri Lanka                               |
| • <b>Achievements:</b> Graduated with Distinction (SGPA: 3.98 / 4.00, A+ Bachelor’s Thesis) |                                         |

SELECTED PUBLICATIONS

- Nadeeshan De Silva, Martin Kellogg, Oscar Chaparro. “**Verifier Warnings Do Not Improve Comprehensibility Prediction**” *International Conference on Evaluation and Assessment in Software Engineering (EASE)*, 2026.
- Nadeeshan De Silva, Martin Kellogg, Oscar Chaparro. “**From Absolute to Relative Code Comprehensibility Prediction**” *arXiv preprint*, 2026.
- Y. Song, J. Mahmud, Nadeeshan De Silva, et al. “**LadyBug: A GitHub Bot for UI-Enhanced Bug Localization in Mobile Apps**” *International Conference on Software Maintenance and Evolution (ICSME)*, 2025.
- J. Mahmud, Nadeeshan De Silva, S. A. Khan, et al. “**On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization**” *International Conference on Software Engineering (ICSE)*, 2024.
- Y. Song, J. Mahmud, Nadeeshan De Silva, et al. “**BURT: A Chatbot for Interactive Bug Reporting.**” *International Conference on Software Engineering (ICSE)*, 2023.
- G. Gamage, Nadeeshan De Silva, et al. “**Phylogenetic tree construction using K-mer forest-based distance calculation**” *International Journal of Online and Biomedical Engineering (iJOE)*, 2023.
- **Best Paper Award:** G. Gamage, Nadeeshan De Silva, et al. “**Alignment-free Whole Genome Comparison Using k-mer Forests**” *International Conference on Advances in ICT for Emerging Regions (ICTer)*, 2019.

HONORS & SERVICE

- **Artifact Evaluation Committee Member**, ASE, ISSTA 2026
- **NSF Travel Grants**, International Conference on Software Engineering (ICSE 2026)
- **Junior Program Committee Member**, International Conference on Mining Software Repositories (MSR) 2025
- **External Reviewer** for ICSE (2023, 2024, 2026, 2027), FSE (2023), ICSME (2023, 2025), ICPC (2025, 2026), IEEE Software (2024)
- **Best Paper Award**, International Conference on Advances in ICT for Emerging Regions (ICTer) 2019
- **ACM SIGSOFT Travel Grants**, International Conference on Software Engineering (ICSE 2024, 2026)
- Presented research at ICSE 2024; Student Volunteer at ICSE 2024, ICSME 2023, ESSEC 2024
- Apache OODT Project Management Committee (PMC) Member & Committer 2021–2023